

Q1 What is the time complexity of Prim's algorithm using an adjacency matrix?

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Your Answer

the time complexity of prim's algorithm using an adjacency matrix $O(V^2)$ and finds the minimum spanning tree(mst) of a weighted graph

Q2 Define the Floyd–Warshall algorithm.

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Your Answer

Floyd- Warshall algorithm is a dynamic programming algorithm used to find the shortest paths between all pairs of vertices in a weighed graphs

Q3 What is Breadth First Search (BFS)?

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Your Answer

breadth first search(BFS) is a graph traversal algorithm that visits vertices level by level, starting from a source vertex

Q4 Which data structure is used in BFS?

Which data structure is used in BFS?

Your Answer

BFS uses a queue (FIFO) data structure and the starting vertex is inserted into the queue and vertices are removed

Q5 What is Depth First Search (DFS)?

What is Depth First Search (DFS)?

Your Answer

depth first search (DFS) is a graph traversal method that explores a vertex as deeply as possible before backtracking

Q6 Which data structure is commonly used in DFS?

2 marks

Which data structure is commonly used in DFS?

Your Answer

DFS commonly uses a stack (LIFO) data structure, either explicitly or through recursion and explores a vertex deeply before returning and visiting other vertices

Q7

A graph contains the vertices A, B, C, D, E, F with edges A-B, A-C, B-D, B-E, C-F, E-F. Starting from A, determine the BFS traversal order. A social network represents users as ve

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Your Answer

start from - A

visit A's neighbours - B,C

visit B's unvisited neighbours -D,E

visit c's unvisited neighbours -F